

Assignment - 3 (Tutorial-3)

1. Given data:

$$P_{g1} = 200 \text{ MW}$$

$$P_{g2} = 400 \text{ MW}$$

$$\text{Speed drop of } P_{g1} = 4\%$$

$$\text{Speed drop of } P_{g2} = 5\%$$

P_{g1} & P_{g2} are parallel operated.

Rated frequency = 50 Hz

Power = 600 MW shared at 50 Hz.

What is the power shared if power is reduced to 400 MW.

(a) To find No load frequencies of generators:

Rated freq. 50 Hz.

G1

$$f_{NL1} \Rightarrow$$

$$s_p = \frac{f_{NL} - f_{rated}}{f_{rated}}$$

$$f_{NL1} = 0.04 \times 50 + 50$$
$$= \underline{\underline{52 \text{ Hz}}}$$

for G2

$$f_{NL2} \Rightarrow f_{NL2} = 0.05 \times 50 + 50$$
$$= \underline{\underline{52.5 \text{ Hz}}}$$

To find the slopes,

G1

$$\text{drop in freq, } \Delta f_1 = 2 \text{ Hz}$$

$$\text{Slope, } m_1 = \frac{P_{g1}}{\Delta f_1} = \frac{200 \text{ MW}}{2 \text{ Hz}}$$
$$m_1 = \underline{\underline{100 \text{ MW/Hz}}}$$

G2

drop in freq: $\Delta f_2 = 2.5 \text{ Hz}$

$$\text{slope, } m_2 = \frac{P_{g2}}{\Delta f_2} \\ = \frac{400 \text{ MW}}{2.5 \text{ Hz}}$$

$$m_2 = \underline{\underline{160 \text{ MW/Hz}}}$$

* To find the new frequency the following power equations are considered;

$$P_{g1(\text{new})} = 100(52 - f_{\text{new}}) \quad \because f_{\text{new}} = \text{new system frequency.}$$

$$P_{g2(\text{new})} = 160(52.5 - f_{\text{new}})$$

The total power after reduction is 400 MW

$$\text{i.e. } P_{g1(\text{new})} + P_{g2(\text{new})} = 400 \text{ MW}$$

$$\Rightarrow 100(52 - f_{\text{new}}) + 160(52.5 - f_{\text{new}}) = 400$$

$$\Rightarrow f_{\text{new}} = \frac{5200 + 52.5 \times 160 - 400}{260}$$

$$\underline{\underline{f_{\text{new}} = 50.77 \text{ Hz}}}$$

$f_{\text{new}} = 50.77 \text{ Hz}$

- ⑥ To find new load frequencies when governors are reset so that 400mw is shared at 50Hz in rated ratio.

$$\text{Rated Ratio} = 200:400 = 1:2$$

$$\text{Total load } (P) = 400\text{mw}$$

$$\text{Share of each generator is } \left(P_{g1} = \frac{1}{3}(P), P_{g2} = \frac{2}{3}(P) \right)$$

$$\text{then, } P_{g1} = \frac{400}{3} = 133.33\text{mw}$$

$$P_{g2} = \frac{2}{3} \times 400 = 266.67\text{mw}$$

Now, new no load frequencies are given by;

① slope is 100mw/Hz ($=m_1$)

$$\text{freq. change, } \Delta f_1 = \frac{133.33\text{mw}}{100\text{mw/Hz}} = \underline{1.33\text{Hz}}$$

$$f_{N1(\text{new})} = 50 + 1.33 = \underline{51.33\text{Hz}}$$

②

slope is 160mw/Hz ($=m_2$)

$$\text{freq. change, } \Delta f_2 = \frac{266.67\text{mw}}{160\text{mw/Hz}} = 1.667\text{Hz}$$

$$f_{N2(\text{new})} = 50 + 1.667 = \underline{51.67\text{Hz}}$$

© Instructions to the microcontroller:

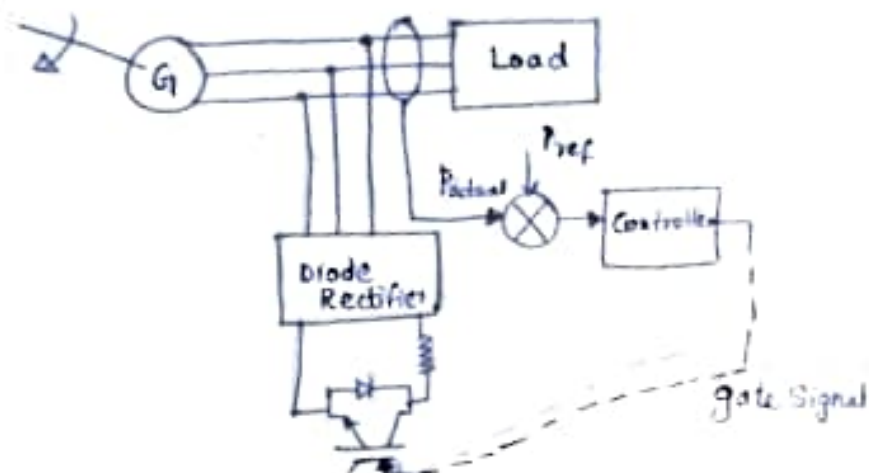
Lookup table:

f is always fixed at 50Hz.

Total load (MW)	P_{g1} (MW)	P_{g2} (MW)	f_{NL1} (Hz)	f_{NL2} (Hz)
600	200	400	52.00	52.50
500	166.67	333.33	51.67	52.083
400	133.33	266.67	51.33	51.67
300	100	200	51.00	51.25

- ② Chopper circuit controls the field excitation/dump load
- * Duty ratio of chopper circuit is; $D = \frac{T_{on}}{T}$

Control circuit Design:



Electronic Load Controller

The total load connected to generator is dump load and actual load. So the total generated power is,

$$P_{gen} = P_{load} + P_{dump}$$

* If P_{load} is decreased;

ie $P_{load} = P_{gen} - P_{dump}$, then,

P_{dump} should increase to balance the equilibrium. As P_{load} decreases the speed and frequency of the generator is increased. Thus, the controller increases the dump load power by varying

the 'D' of the chopper, in case if load increases dump load absorbs less power."

Look-up table :

Actual Load (P.u)	duty of Chopper
1.0	0
0.5	0.5
0	1.0

③ Give data :

If $OS > 115\%$ of rated speed sensor is activated (1)

If $VS \leq 3.5 \text{ mm/s}$ sensor o/p '0'

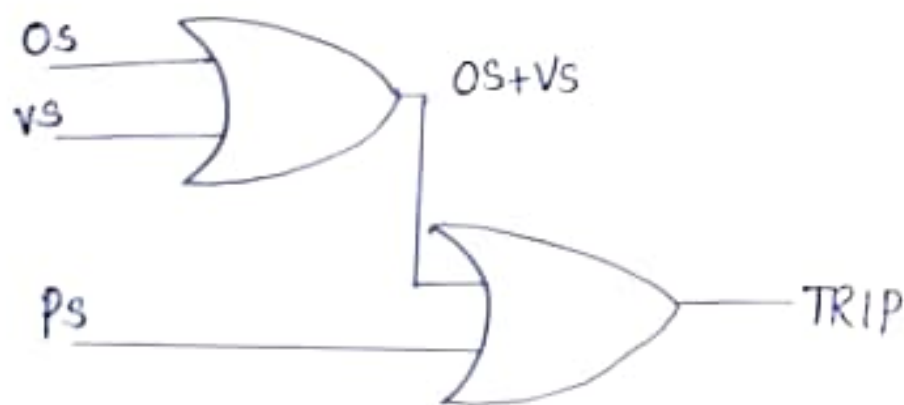
If $PS > 3.0 \text{ bar}$ sensor o/p '1'

TRIP command is executed if any of the sensors are activated.

④ Boolean equation :

$$\boxed{TRIP = OS + VS + PS}$$

⑤ Circuit Implementation (Realization with ^{input} 2 OR gates)



3 input OR gate



OS	VS	PS	TRIP	System state
0	0	0	0	All normal
0	0	1	1	pressure surge causes TRIP
0	1	0	1	Over Vibration causes TRIP
0	1	1	1	Pressure surge and high vibration causing TRIP activation.
1	0	0	1	Over speeding of turbine
1	0	1	1	Pressure surge & overspeed
1	1	0	1	Vibration and overspeed
1	1	1	1	All sensors activated.

- © During monsoon, a grid fault causes sudden removal of load causing overspeeding of turbine. The river water contains debris due to the monsoon causing vibration in the turbine.

Thus the sensor conditions are as follows:

$$OS = 1, VS = 1, \text{ \& } PS = 0.$$

$$\begin{aligned} \text{Then, } TRIP &= OS + VS + PS \\ &= 1 + 1 + 0 = 1 \end{aligned}$$

$$TRIP = 1$$

The protection system is activated immediately and gates of turbine are closed and disconnecting the grid connection. Thus call for an emergency shut down of generator.

(d) If pressure also surges:

$$TRIP = DS + VS + PS$$

$$= 1 + 1 + 1 = 1$$

Emergency shutdown issued by TRIP activation. The generator is now protected by isolating it by keeping gates closed and activation of bypass valves.

Remarks on each scenario:

	Sensor Status	Scenario	Remarks
(1)	0 0 0 0	Normal operation	All parameters are within limits.
2)	0 0 1 1	pressure Surge	penstock pressure exceeds 3.0 bar. <u>causes</u> → load rejection, valve malfunction or water hammer.
3)	0 1 0 1	High Vibration	Bearing Vibration > 3.5 mm/s. <u>causes</u> → Bearing wear, unbalance / debris effect etc
4)	0 1 1 1	Vibration and pressure surge.	Multiple faults occurred; vibration and pressure surge. <u>causes</u> → debris effect etc
5)	1 0 0 1	Over speed	Turbine speed > 575 rpm (ie > 110%) <u>causes</u> → load rejection, grid fault, governor fault.
6)	1 0 1 1	Over speed and pressure	case of load rejection. Turbine accelerates while wicket gates close causing pressure rise

7) 1101

Over speed
and
vibration

Severe Mechanical
Stress;
Causes → load rejection,
bearing failure at high
speed. etc.

8)

1111

All faults

All faults means
complete system failure.
Causes → shaft Breakage or
Penstock rupture.